Fabian A. Alvarez-Primo, Ph.D.

El Paso, TX → Savannah, GA · fpalvarez23@gmail.com · +1 (915) 603-2117 ·
linkedin.com/in/fpalvarez23 · github.com/fabzy4L

Research Interests

[DRAFT — anchor decided 2026-08-26 by working judgment call, pending Fabian's final sign-off] My research program develops manufacturing-grade, quality-controlled biological systems — engineered tissue constructs, iPSC-derived neural and glial cell products, and the computational/data infrastructure to characterize them — for neuropsychiatric and neurodegenerative disease modeling and cell-based therapy. This program bridges three threads from my own training and practice: biomaterials engineering for cell-therapy delivery vehicles (my doctoral work); cGMP-compliant iPSC reprogramming and neural/glial differentiation for psychiatric disease modeling (my postdoctoral work); and process reproducibility, quality-system design, and scale-up discipline drawn from managing large, regulated operations. A connecting method throughout is computational and AI-driven tooling — from critical quality attribute/process parameter extraction for bioprocess data to molecular docking and transcriptomic analysis of psychiatric drug targets and glial biomarkers — applied not as a separate interest but as the instrumentation needed to make neural cell-product manufacturing reproducible, characterizable, and scalable.

Education

Ph.D., Biomedical Materials Science & Engineering — The University of Texas at El Paso, 2016–2020 Dissertation: *"Fabrication, Development, and Characterization of HiPCO SWCNT–Alginate Hydrogel Composites for Cellular Product Applications"* (Advisor: Binata Joddar). Open Access Theses & Dissertations, No. 2922.

B.S., Metallurgical & Materials Engineering — The University of Texas at El Paso, 2010–2015

Relevant coursework: Biomaterials, Bioengineering, Biomedical Engineering, Tissue Engineering, Biomaterials Characterization, Biostatistics, Composites, Alloys, Polymer Science, Instrumentation & Modeling.

Academic & Professional Appointments

Operations Manager II — Amazon, Savannah, GA — *2024–Present* Manage a 60+ associate operations team in a high-volume regulated fulfillment environment (99.7% quality compliance). Built Python-based labor-forecasting and real-time KPI systems; led process-standardization initiatives that improved throughput 82% (220 → 400+ units/hr) and cut recirculation from 5% to <2%. Designed and deployed a three-layer data-intelligence system (browser automation → analytics layer → staffing/rate engine) replacing manual operational workflows. *(Industry role, included to evidence large-scale process/quality-systems leadership directly transferable to biomanufacturing scale-up.)*

Neuropsychiatry Postdoctoral Fellow — Texas Tech University Health Sciences Center El Paso — *May 2022–Nov 2022* † Directed non-integrating Sendai-virus reprogramming (CytoTune-iPS 2.0) of patient PBMCs — isolated via MACS/Lymphoprep density-gradient separation — into clinically

relevant induced pluripotent stem cells (iPSCs) under cGMP-compliant protocols, for differentiation into neuronal/glial lineages and downstream disease modeling. Led biobanking, dissection, and biomarker measurement in postmortem brain tissue (with Southwest Brain Bank / El Paso County Coroner's Office), including primary tissue dissociation (papain/DNase enzymatic digestion) and primary culture establishment (Neurobasal/B-27/bFGF-supplemented neural culture medium) for cryopreserved biobank quality assessment. Owned data structuring and biostatistical analysis (R, Python) across concurrent transcriptomic/genomic (RNA-Seq) studies of neuropsychiatric disease pathophysiology.

Clinical Laboratory Scientist / Manager — West Texas Pain Institute, PLLC, El Paso, TX — *Mar 2021–Apr 2022* Supervised CLIA/COLA/HIPAA-compliant toxicology and pharmacogenomic testing (1,300+ patient samples/month), leading a laboratory team of 9. ‡ Operated and maintained LC-MS/MS, enzyme immunoassay (CEDIA, DRI, QMS, EIA), and RT-PCR/OpenArray pharmacogenomic platforms (48-genotype drug-metabolism panel). Managed LIMS (Stratus DX) and EMR (eClinicalWorks) data pipelines; applied Excel/R/Python analytics to improve DHHS/Drug-of-Abuse protocol workflow.

Science Teacher — Radford School, El Paso, TX — *Jan 2020–Dec 2020* Taught high-school-level Physics (classical mechanics, intro quantum mechanics, solid-state physics), Chemistry (inorganic, organic/biochemistry), AP Environmental/Geological Science, and Biology (ecosystems, evolutionary biology, microbiology, genetics).

Doctoral Research Assistant — Inspired Materials & Stem Cell-Based Tissue Engineering Lab (IMSTEL), UTEP — *Aug 2016–May 2020* Led development and characterization of an electrically conductive cardiovascular tissue composite (HiPco single-walled carbon nanotube–alginate hydrogel) as a cell-therapy delivery vehicle (ASTM/ISO 9001-compliant methodology; collaboration with Marti Lab, Rice University), including zeta-potential characterization of SWNT dispersion/colloidal stability across 11 surfactant systems and bulk electrical conductivity measurement (mS/cm) of crosslinked vs. non-crosslinked composites, identifying the Pluronic F108 formulation as achieving the highest measured conductivity (~2.1 mS/cm crosslinked) — directly informing the electrically conductive scaffold's design for supporting cardiac cell-therapy applications. Contributed to antibacterial wound-healing composites, engineered bone scaffolds, bioprinted integument, a mathematical model of scaffold degradation under physiologic/microgravity conditions, and a protein-based drug-delivery vehicle for neglected tropical disease therapeutics. Mentored 30+ undergraduate/graduate students in electron microscopy and 3D bioprinting; taught the microscopy module of *BME 4371 — Biomaterials Characterization and Biocompatibility*, resulting in a co-authored publication.

Medical Scribe — ProScribe, LLC, El Paso, TX — *Aug 2019–Aug 2020* Clinical documentation support (eClinicalWorks) across multiple outpatient clinics.

Co-Founder / Chief Scientific Officer — Biosvelte, El Paso, TX / Orlando, FL — *Jan 2018–Aug 2020* Co-founded an evidence-based nutraceutical company; led analytical-chemistry quality control for product formulation, manufacturing, and distribution; managed a 10-person sales team.

Chief Executive Officer / Co-Founder — New Financial Solutions, LLC, El Paso, TX — *Jan 2016–Dec 2017* Directed real estate acquisition/resale strategy (El Paso/San Antonio metros); managed a 6-person team and an investor committee.

Sales Representative / Trainer — Alorica (DirecTV/AT&T account), El Paso, TX — *Oct 2016–Feb 2018* Commission-based sales; promoted to trainer, delivering 8 cohort trainings (~160 people) on sales technique and de-escalation.

R&D Engineering Intern — Freeport-McMoRan Climax Molybdenum, Sahuarita, AZ — *May–Aug 2015* Developed ASTM-compliant test methods and SOPs for refractory ceramic (leaching-vessel mortar) durability; conducted compressive-strength, TGA, density, and weight-loss analysis.

Undergraduate Research Assistant — W.M. Keck Center for 3D Innovation (Polymer Extrusion Lab), UTEP — *Nov 2014–May 2015* Synthesized and mechanically characterized (tensile/compression/creep) polymer-composite filaments for 3D printing (biomedical PLA, aerospace ABS/SEBS); XRD/SEM fracture and crystal-structure analysis; contributed to *Wohlers Report 2016* (Appendix E, Materials Properties) in collaboration with the U.S. Air Force and America Makes.

Undergraduate Research Assistant — Materials Research & Technology Institute, UTEP — *May–Aug 2014* Synthesized and characterized bismuth nanoparticles (semiconducting behavior) under Dr. Russell Chianelli, contributing to manuscript preparation. Independently proposed and designed a Co/Ni-promoted MoS■ transition-metal-sulfide catalyst project (hydrothermal and reflux synthesis routes; XRD/SEM/TEM characterization; chelating-agent optimization with thiourea, citric acid, EDTA, and NTA) targeting catalytic hydrodesulphurization of dibenzothiophene for EPA-compliant petroleum desulfurization — proposal-based work supporting a subsequent undergraduate research scholarship application.

Undergraduate Research Assistant — LIMBS International, UTEP — *Jan–Aug 2012* Materials research for low-cost, high-durability prosthetics for pediatric patients in low-resource settings (ethnographic/anthropometric/biomechanical design considerations), under Dr. Roger Gonzalez.

† *CV source documents disagree on this appointment's end date — see Reconciliation Notes.* ‡ *CV source documents disagree on this team size (9 vs. 5) — see Reconciliation Notes.*

Independent Research & Technical Projects

(Solo-developed, post-PhD technical work — included as evidence of independent research/engineering capability, not as funded or deployed systems. Framed honestly per project status below.)

LabOps.AI — Biomanufacturing Process Intelligence Platform *(2026, active development ~7 weeks, prototype)* Solo-built platform applying literature-derived AI extraction to cell/gene-therapy manufacturing data. Built a literature RAG pipeline (bioRxiv/PubMed → Unpaywall → PyMuPDF → Gemini summarization → ChromaDB vector store, ~62 papers ingested) and a 12-table process-data schema seeding 31 Critical Quality Attribute/Critical Process Parameter definitions referencing USP <85>/<1116>, ISO 14644-1, and ICH Q6B standards. FastAPI backend, React/Vite dashboard, instrument adapters for representative bioprocess hardware (Nova BioProfile FLEX, YSI 2950D, Sartorius BIOSTAT); 81 passing automated tests. Local prototype, not clinically or GxP-validated, not externally deployed. GitHub: fabzy4L/LabOps.AI

SERT S438T — Computational Pharmacology of Escitalopram Binding *(May 2026, solo, git-verified)* Independent computational structural biology study of the S438T point mutation in the human serotonin transporter (SERT/*SLC6A4*) and its effect on escitalopram (SSRI) binding. Explicitly framed as originating from the TTUHSCEP postdoctoral iPSC/postmortem-brain transcriptomic work on major depressive disorder. Performed *in-silico* S438T mutagenesis (NERF algorithm, PDB 5I6Z) and multi-seed AutoDock Vina docking (n=5 seeds, exhaustiveness=32); found the rigid-receptor model did not reproduce the experimentally reported 320-fold affinity loss (Andersen et al., 2009) — a well-reasoned negative/nuanced result, with the discrepancy discussed against known limitations of

static docking and molecular dynamics proposed as the next step. Full research article, methods, and computational pipeline documented in-repo. References a forthcoming companion review, *"Cell-Type-Resolved Transcriptomics in Mood and Psychotic Disorders"* (medRxiv, in preparation). GitHub: fabzy4L/sert-s438t-escitalopram

qPCR Expression Atlas Dashboard *(2026, solo, git-verified)* React/Vite dashboard visualizing qPCR gene-expression data (oligodendrocyte/myelin markers — BCAS1, MOBP, MBP, OPALIN, CDH19 — plus CCL2, IGF2, INSRR, SLC27A6, FGF1) comparing control vs. nicotine/vaping-exposed postmortem brain samples. Extends the dataset underlying the Frontiers in Psychiatry publication (see Publications) with independent visualization tooling (heatmap, distribution, and QC-flagging views).

Neuro-Inflammatory Biomarker Analysis in Major Depressive Disorder & Suicidal Behavior *(May 2026, solo, git-verified — no public repo name, internal project)* Statistical/bioinformatics analysis of Luminex multiplex protein-assay data (2 plates, 10 analytes: CNTF, GFAP, NFH, UCHL1, EGF, FGF2, HGF, NGFβ, SCF, VEGF-A) across a clinical cohort (n=70: 35 control, 23 MDD, 12 suicide). Built a full analysis stack — R (normality testing, transformation selection, ANOVA modeling of diagnosis/age/sex/race effects), a FastAPI backend, and a React/TypeScript dashboard. Findings: GFAP significantly associated with age ($p<0.001$); UCHL1 varies significantly by race ($p<0.001$). Self-audited and corrected 8 methodological issues in the original analysis notebook (including flagging that standard curves should use 4-parameter logistic rather than linear regression, with the correct implementation documented) — an explicit, evidenced practice of self-directed methodological rigor.

Image-to-Bioprint Scaffold Pipeline *(June 2026, solo, git-verified)* Python pipeline converting medical imaging data (DICOM MRI/CT or synthetic phantoms) into bioprint-ready STL scaffolds with TPMS (triply periodic minimal surface — gyroid, Schwartz-P, lattice) internal geometry, extending PhD-era bioprinting/tissue-engineering work. Marching-cubes volumetric reconstruction from segmented cross-sections, validated against a reference geometry (Allevi meniscus, 7-point geometric QA suite, 7/7 passing), GelMA/hyaluronic-acid bioink pore-size targeting. Includes device-adjacent regulatory-engineering documentation (ISO 14971 risk register, IEC 62304 compliance notes, RUO labeling) — explicitly research-use-only, not FDA-cleared.

"AI-Enabled Cell & Gene Therapy Biomanufacturing" — Literature Review (in preparation) *(June 2026)* Full draft review manuscript (~7,800 words, 9 sections) synthesizing ~20 primary sources on AI applications and their validation limits in CGT biomanufacturing and AI-guided bioprinting, targeting *Trends in Biotechnology* / *Biotechnology and Bioengineering* / bioRxiv. Not yet submitted. Representative framing: *"the bottleneck is process, not biology"*; cites that 83% of AI-bioprinting studies remain at the lowest validation tier, with none reaching prospective GMP-deployment standards.

Dissertation RAG Pipeline — Retrieval-augmented conversational interface over the author's own doctoral dissertation (font-aware PDF-to-Markdown parsing, ChromaDB, sentence-transformers, Streamlit UI).

BLKPHXLABS AI_BRIEFING — Autonomous biotech-intelligence pipeline (Google Alerts → GCP Cloud Function → Gemini synthesis → TTS/video briefing); serverless GCP architecture, daily automated runs.

Publications

Peer-Reviewed Journal Articles

Vargas-Medrano, J., Carcoba, L. M., Vidal Martinez, G., Mulla, Z. D., Diaz, V., Ruiz-Velasco, A., **Alvarez-Primo, F.**, Colina, G., Iñiguez, S. D., Thompson, P. M., O'Dell, L. E., & Gadad, B. S. (2023). Sex and diet-dependent gene alterations in human and rat brains with a history of nicotine exposure. *Frontiers in Psychiatry, 14*, 1104563. https://doi.org/10.3389/fpsyt.2023.1104563 *(Author position 7 of 12 — from postmortem-brain nicotine-exposure work during the TTUHSCEP postdoctoral appointment.)*

Alvarez-Primo, F., Anil Kumar, S., Manciu, F. S., & Joddar, B. (2019). Fabrication of surfactant-dispersed HiPco single-walled carbon nanotube-based alginate hydrogel composites as cellular products. *International Journal of Molecular Sciences, 20*(19), 4802. *(First author)*

Alonzo, M., Kumar, S. A., Allen, S., Delgado, M., **Alvarez-Primo, F.**, Suggs, L., & Joddar, B. (2020). Hydrogel scaffolds with elasticity-mimicking embryonic substrates promote cardiac cellular network formation. *Progress in Biomaterials, 9*(3), 125–137.

Alonzo, M., Dominguez, E., **Alvarez-Primo, F.**, Quinonez, A., Munoz, E., Puebla, J., et al. (2020). A comparative study in the printability of a bioink and 3D models across two bioprinting platforms. *Materials Letters, 264*, 127382.

Alonzo, M., Primo, F. A., Kumar, S. A., Mudloff, J. A., Dominguez, E., Fregoso, G., et al. (2020). Bone tissue engineering techniques, advances, and scaffolds for treatment of bone defects. *Current Opinion in Biomedical Engineering*, 100248.

Cleetus, C. M., Primo, F. A., Fregoso, G., Raveendran, N. L., Noveron, J. C., Spencer, C. T., et al. (2020). Alginate hydrogels with embedded ZnO nanoparticles for wound healing therapy. *International Journal of Nanomedicine, 15*, 5097.

Carrete, I. A., Bermudez, D., Aguirre, C., **Alvarez-Primo, F. A.**, Anil-Kumar, S., Chinolla, P., et al. (2019). Failure analysis of additively manufactured polyester test specimens exposed to various liquid media. *Journal of Failure Analysis and Prevention, 19*(2), 418–430.

Manuscripts Under Review §

1. **Alvarez-Primo, F.**, & Joddar, B. (2021). Longitudinal stability studies of HiPCO SWCNT–Pluronic F108 suspended alginate hydrogel composite for in vivo applications of cardiac tissue engineering. *Internal Medicine Review*. — Status as of this application: to be confirmed. *(First author)*

Dissertation

1. **Alvarez-Primo, F. A.** (2020). *Fabrication, Development, and Characterization of HiPCO SWCNT–Alginate Hydrogel Composites for Cellular Product Applications.* Doctoral dissertation, The University of Texas at El Paso. Open Access Theses & Dissertations, 2922.

Industry / Technical Reports

1. Wohlers, T., **Alvarez-Primo, F.**, English, J., & Mireles, J. (2016). *Wohlers Report 2016: 3D Printing and Additive Manufacturing State of the Industry — Appendix E: Material Properties.* Wohlers Associates.

Presentations & Invited Lectures

- "Development and characterization of high-purity single-walled carbon nanotube-based gelatin-alginate hydrogel." **Elevator Speech Presentation — 1st Place**, Metallurgy & Materials Engineering Department Day, UTEP, 2018.
- "Fabrication of Surfactant-Dispersed HiPco Single-Walled Carbon Nanotube-Based Alginate Hydrogel Composites as Cellular Products." NSF-PREM Presenter, Materials Research Society National Convention, Boston, MA, Dec 1–6, 2019.
- "Fabrication of Surfactant-Dispersed HiPco Single-Walled Carbon Nanotube-Based Alginate Hydrogel Composites as Cellular Products." Officer Presenter, Alpha Chi National Honor Society National Convention, Albuquerque, NM, Mar 19–21, 2020.

Teaching Experience

The University of Texas at El Paso — *BME 4371, Biomaterials Characterization and Biocompatibility* — Electron Microscopy module instructor (2019); resulted in co-authored publication.

Radford School (El Paso, TX), 2019–2020 — Physics (classical mechanics, intro. physical chemistry, quantum mechanics, solid-state physics); Chemistry (inorganic, intro. organic/biochemistry); AP Environmental/Geological Science; Biology (ecosystems, evolutionary biology, microbiology, genetics); General Science (grades 7–8).

Tutoring (paid & volunteer): Molecular/Cell Biology, Bioengineering, Biochemistry, Anatomy, Algebra through Differential Equations, Corrosion, Introductory Quantum Mechanics, Statistics, Mechanical Behavior of Materials, Physical Metallurgy, Applied Thermochemistry, Mechanics of Materials.

Grants & Fellowships

Award	Amount	Year
Dissertation Completion Fellowship	$16,000 (student stipend)	2019
Eloise & Patrick Wieland Graduate Fellowship (significant work improving human health)	$3,500 (supplies/reagents)	2019
NSF-PREM / Center for Materials Research (Biomedical)	$2,000 (supplies/travel)	2019
The General Property Scholarship for Graduates	$1,000 (stipend)	2018
UTEP MMBME Dept. Student Grant	$500 (stipend)	2018

UTEP Grant for Graduates	$500 (stipend)	2018

Honors & Awards

Tau Beta Pi Engineering Honor Society · Alpha Chi National Honor Society (Alpha Beta Chapter) · Alpha Sigma Mu Materials Honor Society · NSF-PREM Scholar · MRS Scholar/Presenter · Employee of the Month, Alorica (Apr 2017)

Professional Affiliations & Service

- **Society for Biomaterials, UTEP Chapter** — Secretary (2017), Vice-President (2018), **President** (2018–2020, 2 terms)
- American Society for Clinical Pathology (2020)
- Alpha Sigma Mu Honor Society (2018–present)
- Alpha Chi National Honor Society, Alpha Beta Chapter — Professional Chair (2019–2020)
- Tau Beta Pi Honor Society, Texas Theta Chapter (2019–2020)
- Society for Women in Engineering (2015–2020)
- Materials Advantage (2015–2020)
- World Biomaterials Congress — Abstract Revision Committee (2019), as SFB chapter president

Community Service & Outreach

- **Latinos MD** — Co-founder; organized fundraising and MCAT-prep resources for pre-medical students (2020)
- **NSF-PREM / Society for Biomaterials** — Science Fair Judge, Radford School (2019–2020)
- **Alpha Chi Honor Society** — Professional development programming and conference-travel support for students (2019–2020)
- **Society for Biomaterials / Tau Beta Pi** — STEM tutoring and mentoring for BME and multidisciplinary engineering undergraduates (2019–2020)
- **Tau Beta Pi** — Kelly Memorial Food Pantry/YMCA facility restoration volunteer (2019–2020)
- **Society for Women in Engineering** — Non-Newtonian fluids demonstration, Tucson Children's Museum (2014)
- **Gear Up for Engineers** — Tutor/mentor, El Paso High School (2012)

Technical & Laboratory Skills

Cell & Molecular Biology: cGMP-compliant iPSC reprogramming (episomal/viral, non-integrating), cell culture, Western blot, PCR/RT-PCR, DNA ligation/digestion, gel electrophoresis, bacterial transformation, flow cytometry, CRISPR-Cas9, SnapGene, BLAST, Protein Data Bank, VMD (Visual Molecular Dynamics), Avogadro, radioactive uptake assays, live/dead assays, rodent vivarium work.

Biomaterials & Characterization: Tissue engineering, bioprinting, XRD, XPS, XRF, SPR, EDS, DLS, TEM/SEM, Raman/FTIR/NMR spectroscopy, UV-Vis, TGA, DMA, mechanical testing (tensile/compressive/creep), CAD (SolidWorks, Fusion 360, Rhino, AutoCAD), FEA (FEniCS).

Clinical & Regulatory: CLIA/COLA/HIPAA compliance, LC-MS/MS, enzyme immunoassay platforms (CEDIA, DRI, QMS, EIA), DHHS/Drug-of-Abuse protocols, pharmacogenomic testing (RT-PCR/OpenArray), LIMS (Stratus DX), EMR (eClinicalWorks).

Data Science & Computation: Python (Pandas, Scikit-learn, PyTorch), R (Tidyverse, Shiny, Bioconductor), SQL, Power BI, RAG/vector-search architectures (ChromaDB, sentence-transformers), LLM integration (Gemini API), FastAPI, statistical/biostatistical modeling.

Computational Structural Biology: Molecular docking (AutoDock Vina), *in-silico* mutagenesis (NERF algorithm), ligand preparation/conformational optimization (RDKit, OpenBabel), ChimeraX, molecular visualization and structural analysis pipelines.

Languages: English (native/advanced), Spanish (native/advanced).

■ Reconciliation Notes (resolve before finalizing — not for submission)

Cross-referencing the academic CV docx, LinkedIn export, and 2024–2026 resume variants surfaced factual discrepancies. Pick the correct value for each before this CV is submission-ready — search committees can and do check dates:

1. **TTUHSCEP postdoc end date.** LinkedIn: "May 2022–Nov 2022 (7 months)." Academic CV docx: "05/01/2022 – On-going" (never updated after you left). Master 2026 resume: "2022–2023." → I used **May 2022–Nov 2022** as the best-supported value. Confirm actual end date.

2. **WTPI lab team size.** LinkedIn & master resume: "9 people." Academic CV docx: "leading a team of 5 people." → I used **9**. Confirm.

3. **WTPI monthly sample volume.** LinkedIn/master resume: "1,300+." CV docx: "~1200+." → I used **1,300+**. Confirm.

4. **Amazon direct reports.** Master 2026 resume & CLAUDE.md immutable facts: "60+ associates." Fabian_Alvarez_Director_Ops_Biopharma.md: "40+ Direct Reports." → I used **60+** (matches your documented immutable-facts table). Confirm this isn't a "60 total team / 40 direct reports" distinction worth stating precisely.

5. **Publication count claim — UPDATED, mostly resolved.** Your resumes state "8 peer-reviewed publications." With the Frontiers in Psychiatry 2023 paper now added, the actual count of **published, peer-reviewed journal articles is 7** (was 6) — much closer to the claim. Remaining gap: the dissertation (not peer-reviewed) and a 2021 manuscript listed as "under review" in your CV docx (status unknown — it's been 5 years) are still sometimes bundled into the "8" count elsewhere. → Recommend confirming the under-review manuscript's real current status before claiming "8 peer-reviewed" outright; "7 published, 1 under review" is a defensible, fully accurate framing either way.

6. **Research Interests paragraph — RESOLVED (working decision, 2026-08-26), still needs your final sign-off.** After weighing pure biomanufacturing vs. pure neuroscience vs. a blended anchor (you asked to settle this; I made the call under a non-response so we could keep moving), I

locked in the **blended anchor**: manufacturing/quality science for neuropsychiatric and neurodegenerative cell-based therapeutics and disease models. Rationale: it's the only framing every real piece of evidence supports at once — PhD (biomaterials for cell-therapy delivery), postdoc (cGMP iPSC reprogramming *for neuropsychiatric modeling* — the literal bridge), Amazon (quality-systems/scale rigor), and your strongest independent projects on both sides (LabOps.AI + bio-print-pipeline + biomanufacturing-review on manufacturing; SERT/escitalopram + qPCR dashboard + Luminex MDD/suicide study + the Frontiers paper on neuroscience). Pure biomanufacturing was weaker on publication tie and generic against CMaT-adjacent competition; pure neuroscience was thin alone and oversells "neuroengineering" without demonstrated hands-on electrophysiology (still unconfirmed — see note above in conversation). **This is a real 6-year commitment — please explicitly confirm or redirect before the research statement is drafted from it.**

7. **LabOps.AI dates**: git history shows active development April 19–June 2, 2026 (~7 weeks), dormant since. Your Upwork profile and FabianAlvarez_Master_Updated.md list it as "2024–present" — that appears to overstate the timeline based on actual repo evidence. Recommend correcting to reflect real activity dates.

8. **Author-order note (Frontiers in Psychiatry, 2023):** you mentioned your PI placed your name last "out of spite." For the record, the actual published order has you **7th of 12 authors** (senior/corresponding author appears to be Bharathi S. Gadad, listed last) — not literally last. Worth clarifying what you meant (e.g., pushed down from an expected earlier position) so the CV/narrative reflects it accurately if this comes up in an interview.

9. **Electrophysiology/electrical-data claim — FULLY RESOLVED (2026-08-26).** No neuroglia electrophysiology data exists, and none is claimed. The SWBB culture protocol evidences the biochemical/cell-culture side of the postdoc neuroglia work on its own terms. Separately, real electrical/materials data was located and added to the doctoral bullet above: zeta-potential characterization (11 surfactant systems, SWNT dispersion stability — `Zeta Potential Images/`) and bulk conductivity measurements (mS/cm, crosslinked vs. non-crosslinked, PF108 highest at ~2.1 mS/cm — `Corrected Cond.pdf`), both from the IMSTEL doctoral conductive-tissue-composite work. The two projects (postdoc biochemical/neuroglia work vs. doctoral electrical/materials work) remain correctly separated — do not conflate them in the research statement.

10. **MRTI internship purpose — CORRECTED (2026-08-26).** Prior framing described this as "Co/Ni-molybdenum-sulfide catalysts for aqueous toxin chelation." The actual COURI scholarship project description (`Materials Research and Technology Insitute.zip`, Joddar - Dropbox folder) shows the real purpose was catalytic **hydrodesulphurization (HDS) of dibenzothiophene** for EPA-compliant petroleum/fuel desulfurization — chelating agents (thiourea, citric acid, EDTA, NTA) were process reagents to optimize catalyst synthesis/longevity, not an end-goal of "toxin chelation" from water. Corrected in the bullet above. Also note: the source document is a **project proposal/description**, not a results report — frame as proposed/designed work unless separate evidence of completed results and outcomes is found.